Practical no 5

```
CREATE DATABASE online_exam_db;
 USE online_exam_db;

CREATE TABLE customers  customer_id INT
PRIMARY KEY,
customer_name VARCHAR(50));

CREATE TABLE orders (order_id INT
PRIMARY KEY,
 customer_id INT, order_date DATE );

 CREATE TABLE order_items (
item_id INT PRIMARY KEY,
order_id INT,
product_name VARCHAR(50),
 price DECIMAL(10,2)
              );
```

#Insert Data

```
 INSERT INTO customers VALUES
 (1, 'Rahul'),
 (2, 'Sneha'),
 (3, 'Arjun'),
 (4, 'Meena');

 INSERT INTO orders VALUES
 (101, 1, '2025-01-10'),
 (102, 2, '2025-02-15'),
 (103, 1, '2025-03-01');

 INSERT INTO order_items VALUES
 (1, 101, 'Laptop', 50000),
 (2, 101, 'Mouse', 500),
 (3, 102, 'Phone', 20000),
 (4, 103, 'Tablet', 15000),
 (5, 103, 'Headphones', 2000);
```

#INNER JOIN

```
 SELECT c.customer_name, o.order_id
 FROM customers c
 INNER JOIN orders o  ON
```

c.customer_id = o.customer_id;

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_name	order_id
Rahul	101
Sneha	102
Rahul	103

#LEFT JOIN:

```
SELECT c.customer_name, o.order_id
FROM customers c
LEFT JOIN orders o  ON
```

c.customer_id = o.customer_id;

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_name	order_id
Rahul	103
Rahul	101
Sneha	102
Arjun	NULL
Meena	NULL

#RIGHT JOIN:

```
SELECT c.customer_name, o.order_id
FROM customers c
RIGHT JOIN orders o  ON c.customer_id =
o.customer_id;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_name	order_id
Rahul	101
Sneha	102
Rahul	103

#FULL OUTER JOIN

```
SELECT c.customer_name, o.order_id
FROM customers c
LEFT JOIN orders o
ON c.customer_id = o.customer_id

UNION

SELECT c.customer_name, o.order_id
FROM customers c
RIGHT JOIN orders o

ON c.customer_id = o.customer_id;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_name	order_id
Rahul	103
Rahul	101
Sneha	102
Arjun	NULL
Meena	NULL

Result Grid | Form Editor | Field

VIEW

```
#CREATE DATABASE AND TABLES
CREATE DATABASE  company_db;
USE company_db;

CREATE TABLE departments ( dept_id INT
PRIMARY KEY,
dept_name VARCHAR(50)
            );

CREATE TABLE employees (emp_id INT
PRIMARY KEY, emp_name VARCHAR(50),
salary INT, dept_id INT,
 FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
            );

#INSERT DATA
INSERT INTO departments VALUES
(1, 'IT'),
(2, 'HR');
```

```
INSERT INTO employees VALUES
(1, 'Amit', 50000, 1),
(2, 'Neha', 60000, 1),
(3, 'Raj', 40000, 2),
(4, 'Sita', 70000, 1);
```

#CREATE VIEW

```
CREATE VIEW high_salary AS
SELECT emp_name, salary
FROM employees
WHERE salary > 45000;
```

#QUERY VIEW

```
SELECT * FROM high_salary;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	salary
Amit	50000
Neha	60000
Sita	70000

#UPDATE VIEW

```
CREATE OR REPLACE VIEW high_salary AS
SELECT emp_name, salary
FROM employees  WHERE
salary>55000;
```

Output:

SELECT * FROM high_salary;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

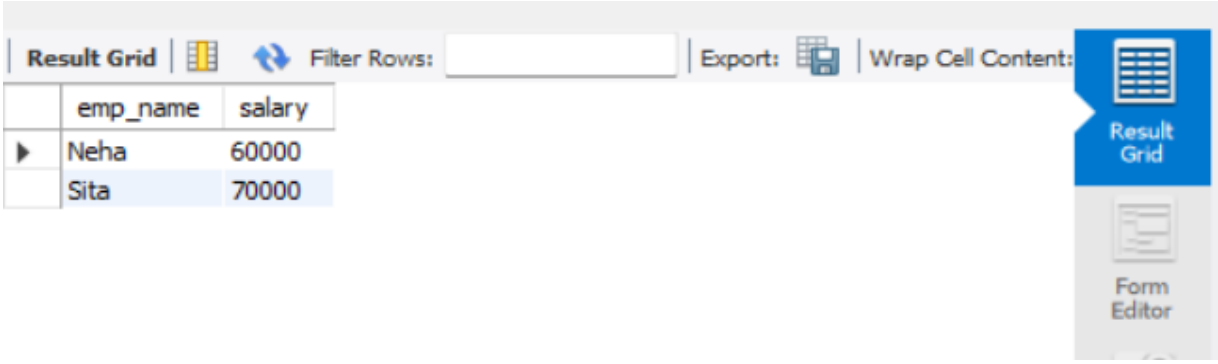
emp_name	salary
Neha	60000
Sita	70000

#SIMPLE VIEW – Employees with salary > 50000 CREATE VIEW high_salary AS

```
SELECT emp_name, salary
FROM employees
WHERE salary > 50000;

SELECT * FROM high_salary;
```

Output:



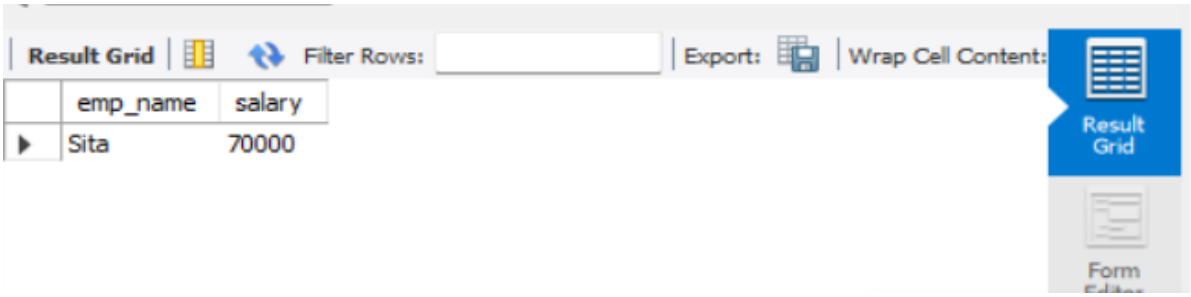
emp_name	salary
Neha	60000
Sita	70000

#Update Simple View: salary > 60000

```
CREATE OR REPLACE VIEW high_salary AS
SELECT emp_name, salary
FROM employees
WHERE salary > 60000;

SELECT * FROM high_salary;
```

Output After Update**:**



emp_name	salary
Sita	70000

#Drop Simple View
```
DROP VIEW high_salary;
```

COMPLEX VIEW – Department Summary

```
CREATE VIEW dept_summary AS
SELECT d.dept_name,
COUNT(e.emp_id) AS total_employees,
SUM(e.salary) AS total_salary
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
 GROUP BY d.dept_name;

SELECT * FROM dept_summary;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	total_employees	total_salary
IT	3	180000
HR	1	40000

Result Grid | Form Editor

#Update Complex View: Add average salary

```
CREATE OR REPLACE VIEW dept_summary AS
SELECT d.dept_name,
COUNT(e.emp_id) AS total_employees,
SUM(e.salary) AS total_salary,
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;

SELECT * FROM dept_summary;
```

Output :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	total_employees	total_salary
IT	3	180000
HR	1	40000

Result Grid | Form Editor

#Drop Complex View

```
DROP VIEW dept_summary;
```

#MATERIALIZED VIEW

#Create table to simulate Materialized View

```
CREATE TABLE emp_summary AS
SELECT dept_id,
COUNT(emp_id) AS total_employees,
SUM(salary) AS total_salary
FROM employees
GROUP BY dept_id;

SELECT * FROM emp_summary;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	total_employees	total_salary
1	3	180000
2	1	40000

Result Grid

Form Editor

#Refresh Materialized View

```
TRUNCATE TABLE emp_summary;
INSERT INTO emp_summary
SELECT dept_id,
COUNT(emp_id),
SUM(salary)
FROM employees
GROUP BY dept_id;

SELECT * FROM emp_summary;
```

#Drop Materialized View

```
DROP TABLE emp_summary;
```

#DROP VIEW

```
DROP VIEW high_salary;
```